

# WHY ABORTION?

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HOW DO ATTITUDES TOWARD ABORTION VARY CROSS NATIONALLY, AND TO WHAT EXTENT ARE THESE ATTITUDES RELATED TO PRO NATALIST VIEWS AND HOUSEHOLD FINANCIAL SATISFACTION?



- ❖ Moral theology & ethical dilemmas
- ❖ Definitions of life steps and personal choices
- ❖ Gender roles & autonomy
- ❖ Sexual morality & norms
- ❖ Political ideologies & polarization

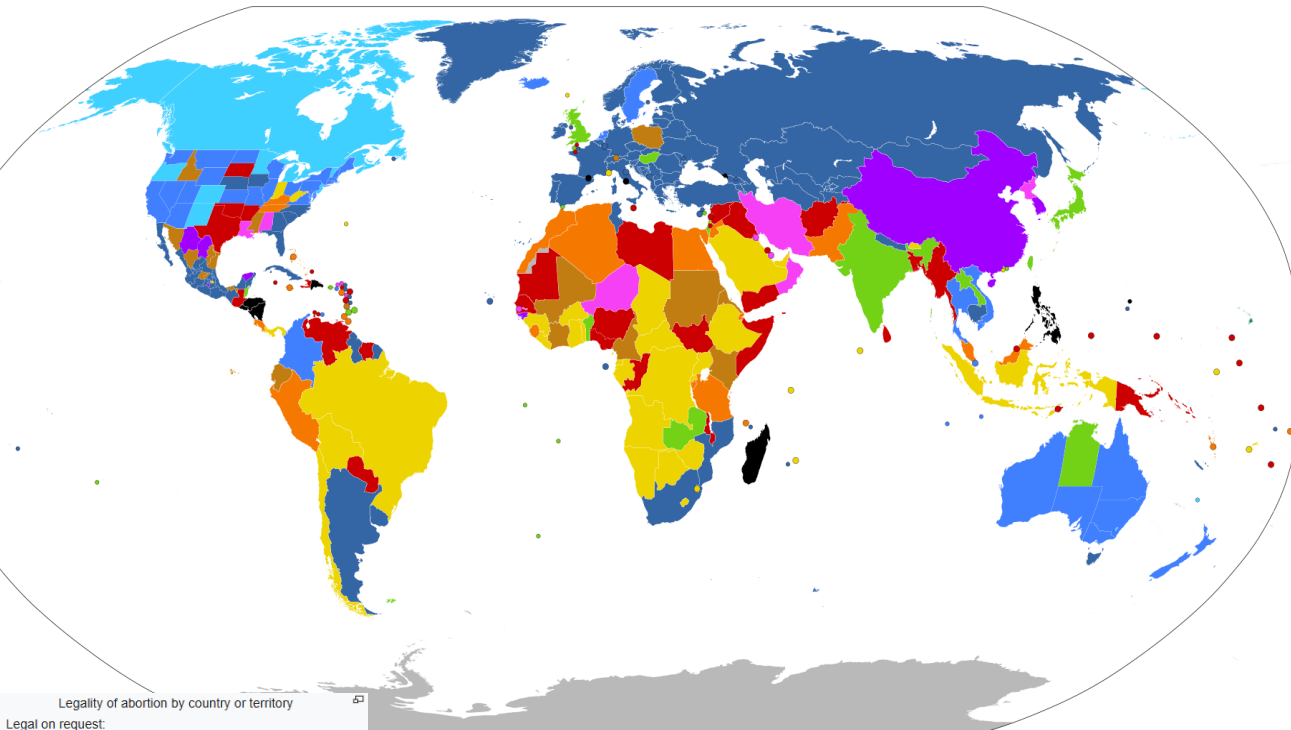
# WHICH FRAMEWORK



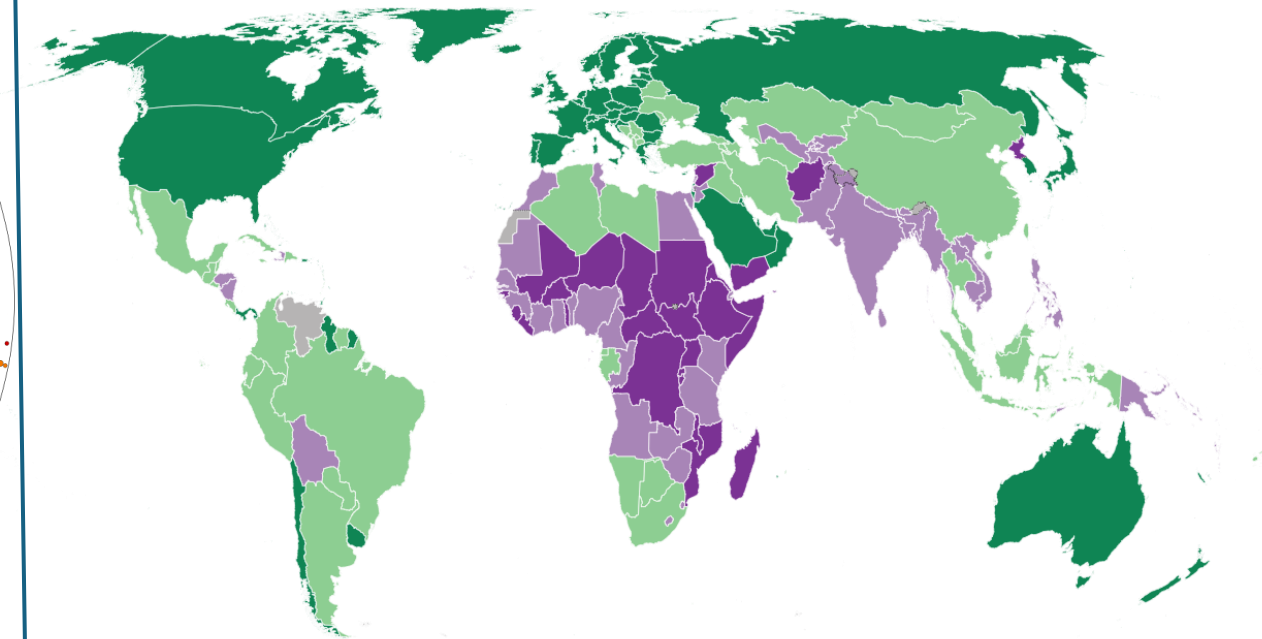
## Main three frameworks:

- ❖ Economic Uncertainty Theory
- ❖ Pronatalism and Demographic Anxiety
- ❖ Gender Inequality Framework

# RESEARCH AIMS & COMPARISON



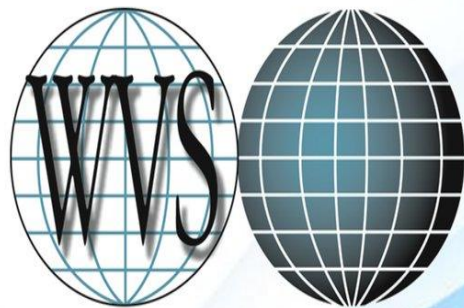
Source: [https://en.wikipedia.org/wiki/Abortion\\_law](https://en.wikipedia.org/wiki/Abortion_law)



Source: World Bank Group



# DATA AND VARIABLES

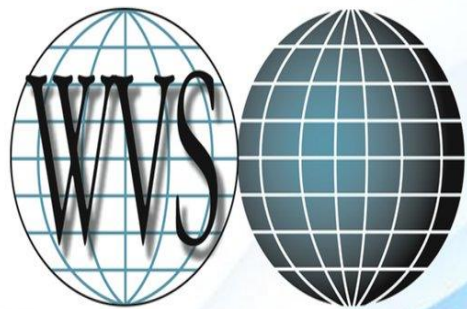


**WVS wave 7 (2017-2020) data release on**

**2007 2020**

|                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Dependent variable:</b>                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>Q184: Please tell me for each of the following statements whether you think it can always be justified, never be justified, or something in between, using this card. Abortion</li> </ul>                                                                                                                                          | 1.- Never justifiable 2.- 2 3.- 3 4.- 4 5.- 5 6.- 6 7.- 7 8.- 8 9.- 9 10.- Always justifiable -1.- Don't know -2.- No answer -4.- Not asked in survey -5.- Missing; Not available                                                                                                                                                     |
| <b>Independent variables include:</b>                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>Demographics</li> </ul>                                                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>Q262: Age at time of interview</li> <li>Q261: birth year</li> <li>Q260: Respondent's sex</li> <li>IncomeWB: Income group country (Low, lower-middle, upper-middle, high) [World Bank, 2019] <a href="https://data.worldbank.org/indicator">https://data.worldbank.org/indicator</a></li> </ul> |
| <ul style="list-style-type: none"> <li>Q273. Are you currently (read out and code one answer only): Marital status</li> </ul>                                                                                                                                                                                                                                             | 1 Married/ 2 Living together as married/ 3 Divorced/ 4 Separated/ 5 Widowed/ 6 Single                                                                                                                                                                                                                                                 |
| <ul style="list-style-type: none"> <li>H_URBRURAL:Urban-Rural: Urban-Rural Settlement type</li> </ul>                                                                                                                                                                                                                                                                     | 1.- Urban 2.- Rural -5.- No answer; Missing -4.- Not asked                                                                                                                                                                                                                                                                            |
| GII: Gender Inequality Index                                                                                                                                                                                                                                                                                                                                              | (GII)(0 to 1)[UNDP, 2018] <a href="http://hdr.undp.org/en/data">http://hdr.undp.org/en/data</a> :<br>GII is a composite metric of gender inequality using three dimensions: reproductive health, empowerment and the labour market. A low GII value indicates low inequality between women and men, and vice-versa.                   |
| <ul style="list-style-type: none"> <li>Q37: Duty towards society to have children: Do you agree, disagree or neither agree nor disagree with the following statements? –It is a duty towards society to have children</li> </ul>                                                                                                                                          | 1.- Agree strongly 2.- Agree 3.- Neither agree nor disagree 4.- Disagree 5.- Disagree strongly -1.- Don't know -2.- No answer -4.- Not asked -5.- Missing; Unknown                                                                                                                                                                    |
| <ul style="list-style-type: none"> <li>Q50 Satisfaction with financial situation of household: How satisfied are you with the financial situation of your household? If '1' means you are completely dissatisfied on this scale, and '10' means you are completely satisfied, where would you put your satisfaction with your household's financial situation?</li> </ul> | 10.- Satisfied 9.- 9 8.- 8 7.- 7 6.- 6 5.- 5 4.- 4 3.- 3 2.- 2 1.- Dissatisfied -1.- Don't know -2.- No answer -4.- Not asked -5.- Missing; Unknown                                                                                                                                                                                   |

# MISSING VALUES



**WVS wave 7 (2017-2020) data release on**

**2007 2020**

- ❖ Not a good way to show the data
- ❖ Bayesian model-based imputation?
- ❖ Model-based Bayesian imputation?

| Variable   | Type    | N     | Missing | Mean    | Median  | SD      | Min   | Max     |
|------------|---------|-------|---------|---------|---------|---------|-------|---------|
| Q184       | numeric | 97220 | 0       | 3.43    | 2.00    | 3.06    | -5    | 10.00   |
| I_ABORTLIB | numeric | 94945 | 2275    | 0.28    | 0.11    | 0.33    | 0     | 1.00    |
| Q262       | numeric | 97220 | 0       | 42.93   | 41.00   | 16.88   | -5    | 103.00  |
| Q261       | numeric | 97220 | 0       | 1943.38 | 1977.00 | 251.18  | -5    | 2007.00 |
| Q260       | numeric | 97220 | 0       | 1.52    | 2.00    | 0.53    | -5    | 2.00    |
| incomeWB   | numeric | 96773 | 447     | 3.06    | 3.00    | 0.84    | 1     | 4.00    |
| Q273       | numeric | 97220 | 0       | 2.63    | 1.00    | 2.19    | -5    | 6.00    |
| H_URBRURAL | numeric | 97220 | 0       | 1.32    | 1.00    | 0.48    | -5    | 2.00    |
| GII        | numeric | 96773 | 447     | -794.18 | 0.30    | 2704.29 | -9999 | 0.55    |
| Q37        | numeric | 97220 | 0       | 2.71    | 3.00    | 1.38    | -5    | 5.00    |
| Q50        | numeric | 97220 | 0       | 6.16    | 6.00    | 2.51    | -5    | 10.00   |
| Q274       | numeric | 97220 | 0       | 1.49    | 2.00    | 2.10    | -5    | 24.00   |

Where are the missing values?

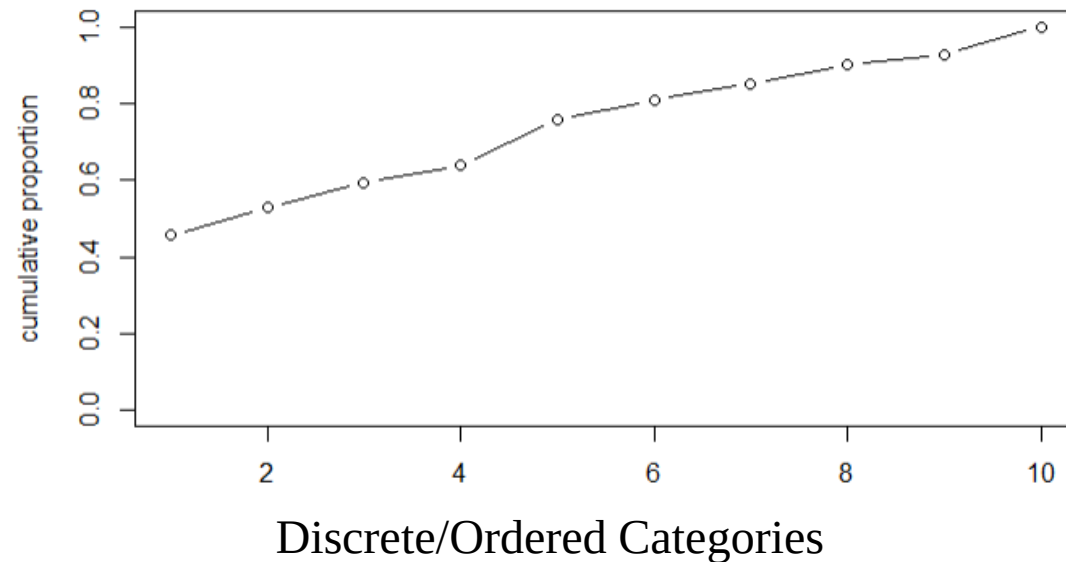
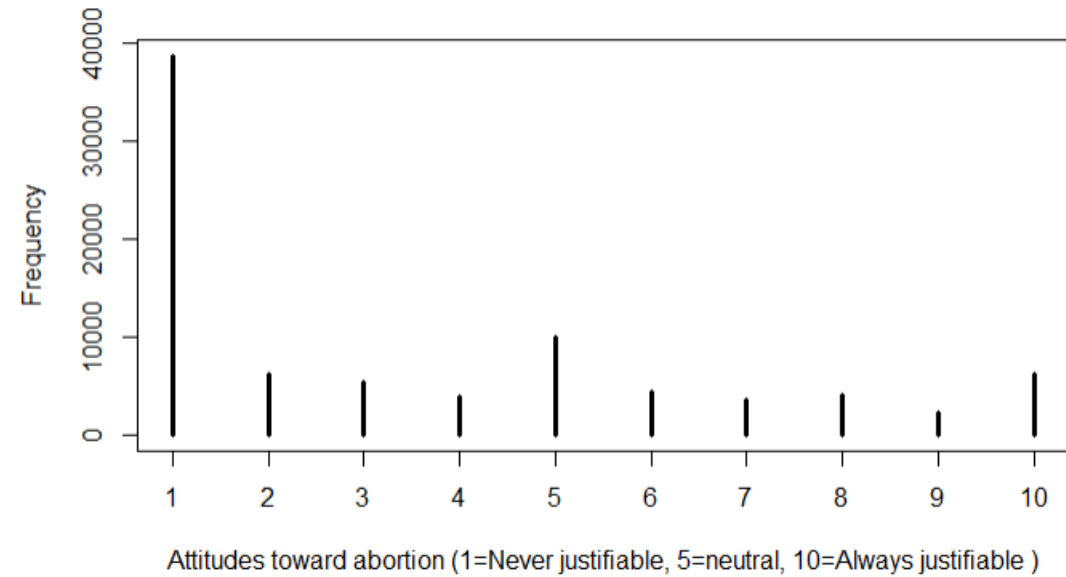
After simple cleaning:

| Complete Variable Summary |         |       |         |       |        |       |       |        |        |
|---------------------------|---------|-------|---------|-------|--------|-------|-------|--------|--------|
| Variable                  | Type    | N     | Missing | Mean  | Median | SD    | Min   | Max    | Levels |
| abortion_justif           | numeric | 84801 | 0       | 3.54  | 2.00   | 3.00  | 1.00  | 10.00  | NA     |
| age                       | numeric | 84801 | 0       | 42.93 | 41.00  | 16.43 | 16.00 | 103.00 | NA     |
| sex                       | numeric | 84801 | 0       | 1.52  | 2.00   | 0.50  | 1.00  | 2.00   | NA     |
| marital_status            | numeric | 84801 | 0       | 2.63  | 1.00   | 2.13  | 1.00  | 6.00   | NA     |
| urban_rural               | numeric | 84801 | 0       | 1.34  | 1.00   | 0.47  | 1.00  | 2.00   | NA     |
| duty_children             | numeric | 84801 | 0       | 2.74  | 3.00   | 1.28  | 1.00  | 5.00   | NA     |
| fin_satisfaction          | numeric | 84801 | 0       | 6.21  | 6.00   | 2.44  | 1.00  | 10.00  | NA     |
| country                   | numeric | 84801 | 0       | 2.99  | 3.00   | 0.83  | 1.00  | 4.00   | NA     |
| GII                       | numeric | 84801 | 0       | 0.30  | 0.32   | 0.16  | 0.04  | 0.55   | NA     |

## DISTRIBUTED OUTCOME



Whether you think it can always be justified, never be justified, or something in between, using this card.





## MODEL AND PRIORS



Fixed effect:

$\text{abortion\_justif}_i \sim \text{OrderedLogistic}(\phi_i, \alpha)$

$$\begin{aligned} \phi_i = & \beta_1 \text{ duty\_children}_i + \beta_2 \text{ fin\_satisfaction}_i + \beta_3 \text{ age}_i + \beta_4 \text{ sex}_i \\ & + \beta_5 \text{ marital\_status}_i + \beta_6 \text{ urban\_rural}_i + \beta_7 \text{ GII}_i + \beta_8 \text{ country}_i \end{aligned}$$

$\beta_j \sim \text{Normal}(0, 0.5)$

$\alpha_k \sim \text{Normal}(0, 0.5)$

# POSTERIOR SUMMARY OF REGRESSION COEFFICIENTS



Regression Coefficients:

|                  | Estimate | Est.Error | l-89% CI | u-89% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept        | 2.30     | 0.07      | 2.19     | 2.40     | 1.00 | 4316     | 3330     |
| duty_children    | 0.33     | 0.01      | 0.32     | 0.33     | 1.00 | 7289     | 3121     |
| fin_satisfaction | -0.00    | 0.00      | -0.00    | 0.00     | 1.00 | 8586     | 3034     |
| age              | -0.00    | 0.00      | -0.01    | -0.00    | 1.00 | 4543     | 3582     |
| sex              | -0.06    | 0.01      | -0.08    | -0.03    | 1.00 | 5767     | 2820     |
| marital_status   | 0.03     | 0.00      | 0.02     | 0.03     | 1.00 | 9092     | 3092     |
| urban_rural      | -0.17    | 0.01      | -0.19    | -0.15    | 1.00 | 5844     | 2816     |
| GII              | -4.23    | 0.07      | -4.33    | -4.12    | 1.00 | 3580     | 3288     |
| country          | 0.07     | 0.01      | 0.05     | 0.09     | 1.00 | 3677     | 3364     |

| Parameter        | Median | 95% CI        | pd     | Rhat  | ESS     |
|------------------|--------|---------------|--------|-------|---------|
| Intercept        | 9.96   | [8.73, 11.31] | 100%   | 0.999 | 4241.00 |
| duty_children    | 1.39   | [1.37, 1.40]  | 100%   | 1.000 | 7219.00 |
| fin_satisfaction | 1.00   | [0.99, 1.00]  | 58.50% | 1.000 | 8374.00 |
| age              | 1.00   | [0.99, 1.00]  | 100%   | 1.000 | 4598.00 |
| sex              | 0.95   | [0.92, 0.97]  | 99.98% | 0.999 | 5660.00 |
| marital_status   | 1.03   | [1.02, 1.04]  | 100%   | 1.000 | 8795.00 |
| urban_rural      | 0.84   | [0.82, 0.87]  | 100%   | 1.000 | 5791.00 |
| GII              | 0.01   | [0.01, 0.02]  | 100%   | 1.000 | 3541.00 |
| country          | 1.07   | [1.04, 1.09]  | 100%   | 0.999 | 3667.00 |



# POSTERIOR SUMMARY OF REGRESSION COEFFICIENTS



Regression Coefficients:

|                  | Estimate | Est.Error | l-89% CI | u-89% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept        | 2.30     | 0.07      | 2.19     | 2.40     | 1.00 | 4316     | 3330     |
| duty_children    | 0.33     | 0.01      | 0.32     | 0.33     | 1.00 | 7289     | 3121     |
| fin_satisfaction | -0.00    | 0.00      | -0.00    | 0.00     | 1.00 | 8586     | 3034     |
| age              | -0.00    | 0.00      | -0.01    | -0.00    | 1.00 | 4543     | 3582     |
| sex              | -0.06    | 0.01      | -0.08    | -0.03    | 1.00 | 5767     | 2820     |
| marital_status   | 0.03     | 0.00      | 0.02     | 0.03     | 1.00 | 9092     | 3092     |
| urban_rural      | -0.17    | 0.01      | -0.19    | -0.15    | 1.00 | 5844     | 2816     |
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| Parameter        | Median | 95% CI        | pd     | Rhat  | ESS     |
|------------------|--------|---------------|--------|-------|---------|
| Intercept        | 9.96   | [8.73, 11.31] | 100%   | 0.999 | 4241.00 |
| duty_children    | 1.39   | [1.37, 1.40]  | 100%   | 1.000 | 7219.00 |
| fin_satisfaction | 1.00   | [0.99, 1.00]  | 58.50% | 1.000 | 8374.00 |
| age              | 1.00   | [0.99, 1.00]  | 100%   | 1.000 | 4598.00 |
| sex              | 0.95   | [0.92, 0.97]  | 99.98% | 0.999 | 5660.00 |
| marital_status   | 1.03   | [1.02, 1.04]  | 100%   | 1.000 | 8795.00 |
| urban_rural      | 0.84   | [0.82, 0.87]  | 100%   | 1.000 | 5791.00 |
| GII              | 0.01   | [0.01, 0.02]  | 100%   | 1.000 | 3541.00 |
| country          | 1.07   | [1.04, 1.09]  | 100%   | 0.999 | 3667.00 |

## PREDICTION SCENARIOS

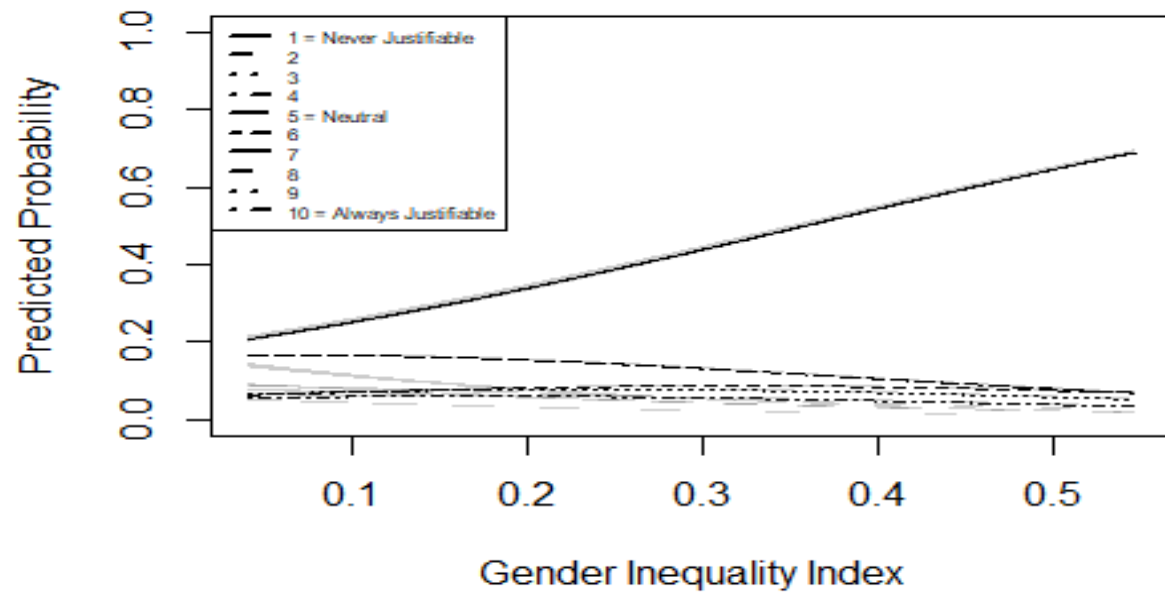
```
cases <- data.frame(
  duty_children = c(1, 3, 5),
  fin_satisfaction = c(1, 5, 10),
  age = mean(dropMissing_data$age),
  sex = 1,          # 1 = Male
  marital_status = 1, # 1 = Married
  urban_rural = 1,   # 1 = Urban
  GII = mean(dropMissing_data$GII),
  country=1 #Low income
)
cases
```

```
pred_abortion <- predict(fit_abortion,
  newdata = cases)
pred_abortion
```

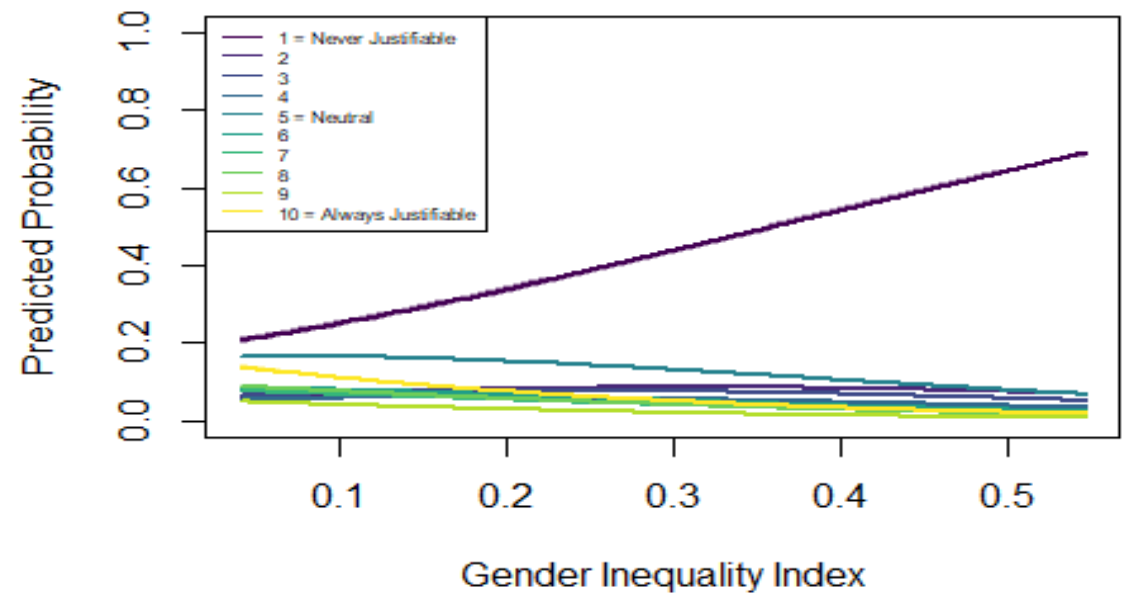
| Duty to Have Children | Financial Satisfaction | P(Y=1) | P(Y=2) | P(Y=3) | P(Y=4) | P(Y=5) | P(Y=6) | P(Y=7) | P(Y=8) | P(Y=9) | P(Y=10) |
|-----------------------|------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|---------|
| 1 (strong pronatal)   | 1 (low)                | 0.590  | 0.080  | 0.064  | 0.044  | 0.100  | 0.039  | 0.022  | 0.024  | 0.012  | 0.026   |
| 3 (moderate)          | 5 (medium)             | 0.454  | 0.089  | 0.075  | 0.053  | 0.131  | 0.051  | 0.041  | 0.039  | 0.018  | 0.050   |
| 5 (low pronatal)      | 10 (high)              | 0.298  | 0.076  | 0.078  | 0.065  | 0.152  | 0.073  | 0.063  | 0.069  | 0.039  | 0.090   |

## VISUALIZATION MODELS

**Predicted Abortion Justifiability by GII**



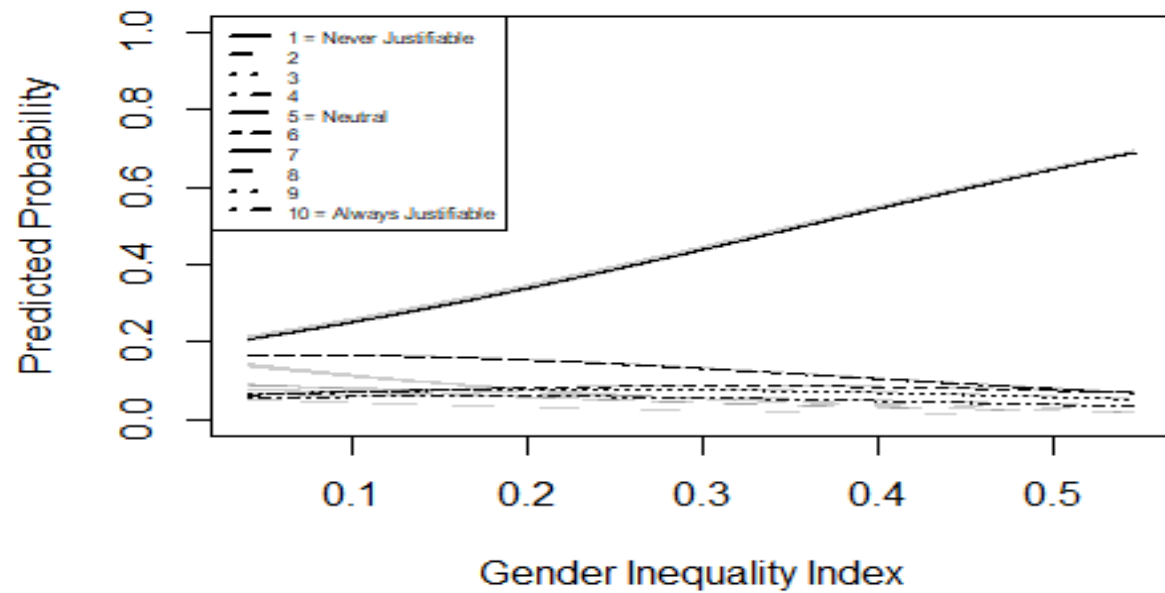
**Predicted Abortion Justifiability by GII**



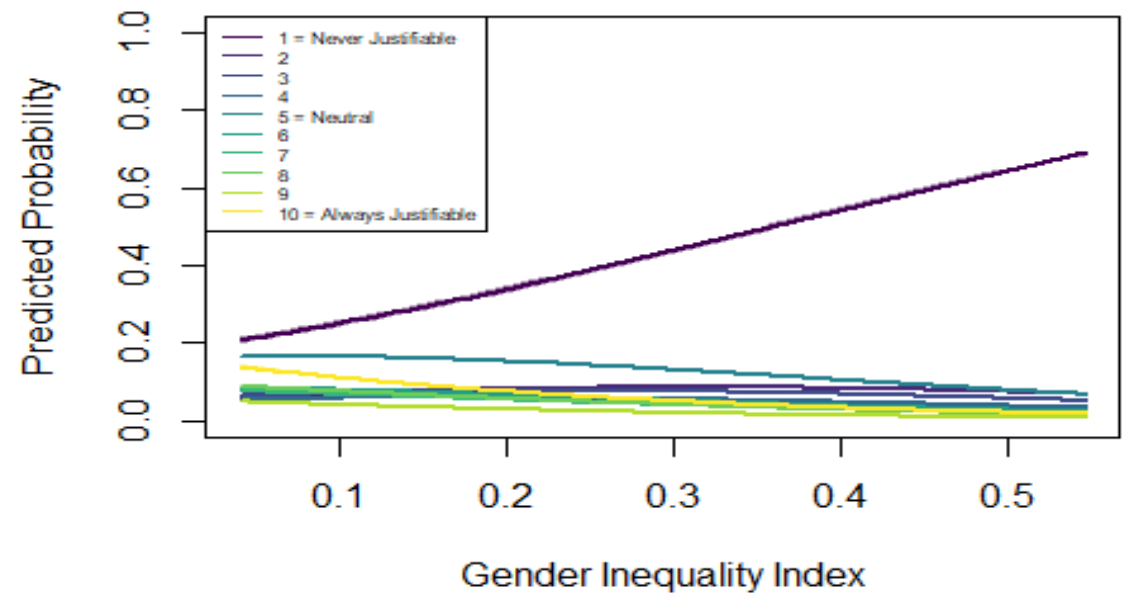


## VISUALIZATION MODELS

**Predicted Abortion Justifiability by GII**



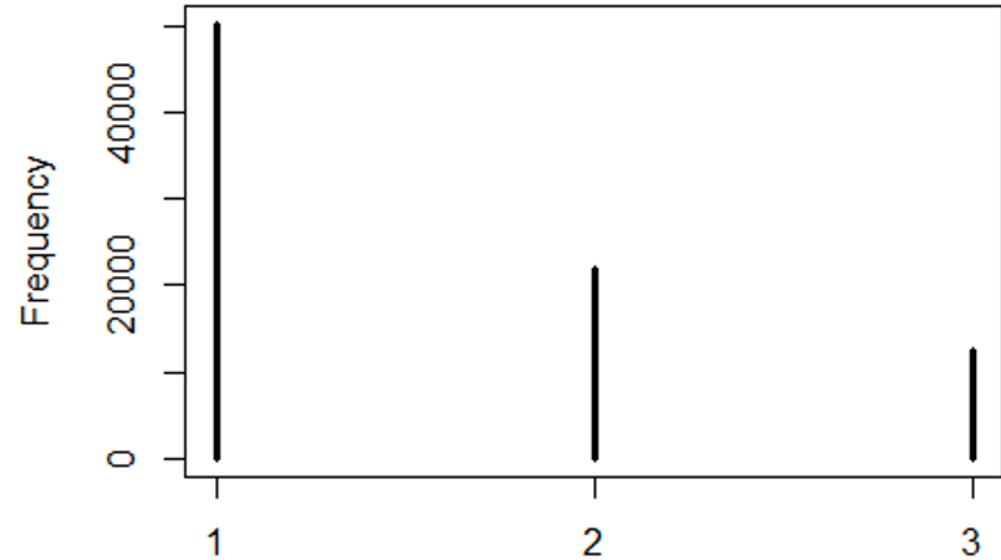
**Predicted Abortion Justifiability by GII**



## RECODED ABORTION ATTITUDES (3 CATEGORIES)

1= Low justifiability , 2=Neutral, 3= High justifiability

```
dropMissing_data_c <- dropMissing_data %>%  
  mutate(abortion_justif_3cat = case_when(  
    abortion_justif %in% 1:3 ~ 1,  
    abortion_justif %in% 4:7 ~ 2,  
    abortion_justif %in% 8:10 ~ 3  
  ))
```



## NEW MODEL AND CATEGORIES

Regression Coefficients (3 CATEGORIES) :

|                  | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept[1]     | 0.31     | 0.07      | 0.16     | 0.46     | 1.00 | 2480     | 2825     |
| Intercept[2]     | 1.97     | 0.08      | 1.82     | 2.11     | 1.00 | 2553     | 2901     |
| duty_children    | 0.35     | 0.01      | 0.34     | 0.37     | 1.00 | 3827     | 2671     |
| fin_satisfaction | 0.00     | 0.00      | -0.00    | 0.01     | 1.00 | 5427     | 2994     |
| age              | -0.00    | 0.00      | -0.01    | -0.00    | 1.00 | 3576     | 2736     |
| sex              | -0.05    | 0.01      | -0.08    | -0.02    | 1.00 | 3497     | 2600     |
| marital_status   | 0.04     | 0.00      | 0.03     | 0.04     | 1.00 | 5326     | 3141     |
| urban_rural      | -0.18    | 0.02      | -0.22    | -0.15    | 1.00 | 3301     | 2459     |
| GII              | -4.01    | 0.08      | -4.16    | -3.87    | 1.00 | 2341     | 2118     |
| country          | 0.17     | 0.01      | 0.14     | 0.20     | 1.00 | 2455     | 2118     |

Regression Coefficients (10 CATEGORIES) :

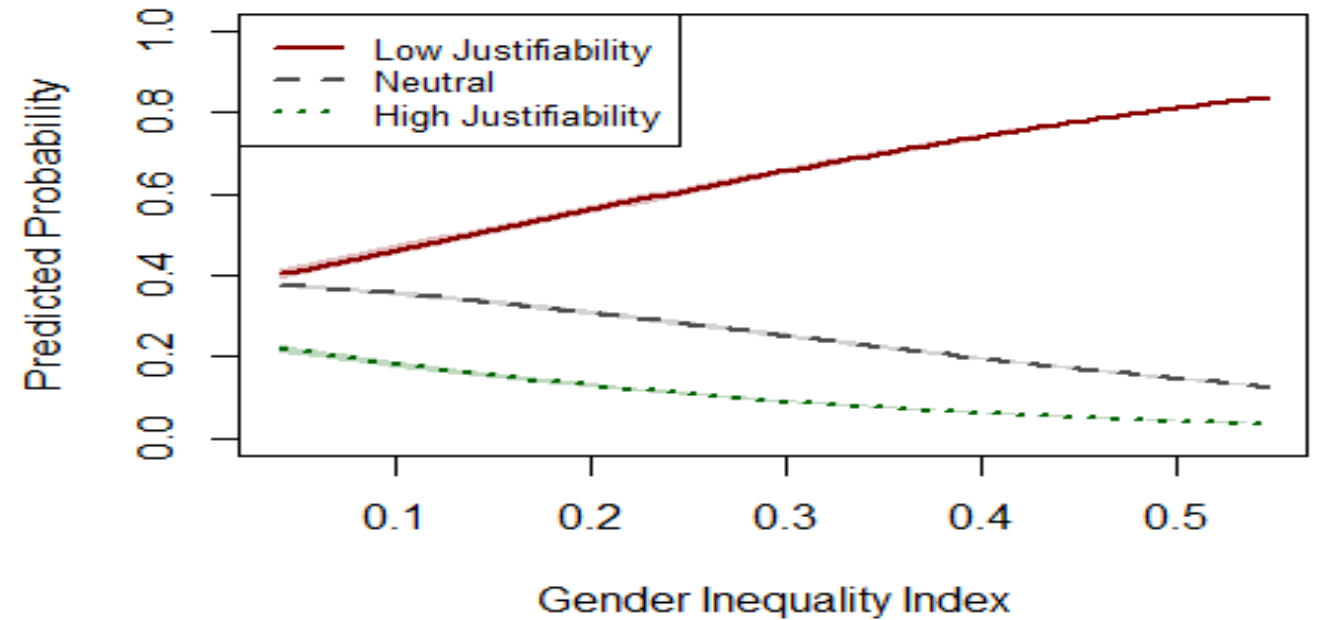
|                  | Estimate | Est.Error | l-89% CI | u-89% CI | Rhat | Bulk_ESS | Tail_ESS |
|------------------|----------|-----------|----------|----------|------|----------|----------|
| Intercept        | 2.30     | 0.07      | 2.19     | 2.40     | 1.00 | 4316     | 3330     |
| duty_children    | 0.33     | 0.01      | 0.32     | 0.33     | 1.00 | 7289     | 3121     |
| fin_satisfaction | -0.00    | 0.00      | -0.00    | 0.00     | 1.00 | 8586     | 3034     |
| age              | -0.00    | 0.00      | -0.01    | -0.00    | 1.00 | 4543     | 3582     |
| sex              | -0.06    | 0.01      | -0.08    | -0.03    | 1.00 | 5767     | 2820     |
| marital_status   | 0.03     | 0.00      | 0.02     | 0.03     | 1.00 | 9092     | 3092     |
| urban_rural      | -0.17    | 0.01      | -0.19    | -0.15    | 1.00 | 5844     | 2816     |
| GII              | -4.23    | 0.07      | -4.33    | -4.12    | 1.00 | 3580     | 3288     |
| country          | 0.07     | 0.01      | 0.05     | 0.09     | 1.00 | 3677     | 3364     |



## Predicted Abortion Attitudes by Gender Inequality



## Predicted Abortion Justifiability by GII (3 Categories)



- ❖ GII was chosen because it captures structural gender inequality across countries.
- ❖ It showed the strongest effect size and most consistent association in the model.
- ❖ Higher GII : more people say abortion is never justifiable.

## SECOND MODEL AND PRIORS



Fixed effect:

$\text{abortion\_justif}_i \sim \text{OrderedLogistic}(\phi_i, \alpha)$

$\phi_i = \beta_1 \text{ duty\_children}_i + \beta_2 \text{ fin\_satisfaction}_i + \beta_3 \text{ age}_i + \beta_4 \text{ sex}_i$   
 $+ \beta_5 \text{ marital\_status}_i + \beta_6 \text{ urban\_rural}_i + \beta_7 \text{ GII}_i + \beta_8 \text{ country}_i$

$\beta_j \sim \text{Normal}(0, 0.5)$

$\alpha_k \sim \text{Normal}(0, 0.5)$

Random intercepts for country:

$\text{abortion\_justif}_i \sim \text{OrderedLogistic}(\phi_i, \alpha)$

$\phi_i = \beta_1 \text{ duty\_children}_i + \beta_2 \text{ fin\_satisfaction}_i + \beta_3 \text{ age}_i + \beta_4 \text{ sex}_i$   
 $+ \beta_5 \text{ marital\_status}_i + \beta_6 \text{ urban\_rural}_i + \beta_7 \text{ GII}_i + (1 \mid \text{country}_i)$

$\beta_j \sim \text{Normal}(0, 2)$

$\alpha_k \sim \text{Normal}(0, 5)$

$\sigma_{\text{country}} \sim \text{Exponential}(1)$

## SUMMARY OF THE SECOND ONE

| Parameter              | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat |
|------------------------|----------|-----------|----------|----------|------|
| Intercept[1]           | -0.15    | 0.08      | -0.31    | 0.01     | 1    |
| Intercept[2]           | 0.21     | 0.08      | 0.04     | 0.36     | 1    |
| Intercept[3]           | 0.53     | 0.08      | 0.36     | 0.68     | 1    |
| Intercept[4]           | 0.77     | 0.08      | 0.6      | 0.93     | 1    |
| Intercept[5]           | 1.46     | 0.08      | 1.29     | 1.62     | 1    |
| Intercept[6]           | 1.82     | 0.08      | 1.66     | 1.98     | 1    |
| Intercept[7]           | 2.19     | 0.08      | 2.02     | 2.35     | 1    |
| Intercept[8]           | 2.69     | 0.08      | 2.52     | 2.85     | 1    |
| Intercept[9]           | 3.06     | 0.08      | 2.89     | 3.22     | 1    |
| duty_children          | 0.33     | 0.01      | 0.31     | 0.34     | 1    |
| fin_satisfaction       | 0.01     | 0         | 0        | 0.01     | 1    |
| age                    | 0.01     | 0         | 0.01     | 0.02     | 1    |
| sex                    | -0.05    | 0.01      | -0.08    | -0.03    | 1    |
| marital_status         | 0        | 0.01      | -0.01    | 0.02     | 1    |
| urban_rural            | -0.21    | 0.01      | -0.24    | -0.18    | 1    |
| GII                    | -3.32    | 0.12      | -3.56    | -3.09    | 1    |
| sd(Intercept)[country] | 0.10     | 0         | 0.09     | 0.11     | 1    |

**$\text{abortion\_justif}_i \sim \text{OrderedLogistic}(\phi_i, \alpha)$**

**$\phi_i = \beta_1 \text{ duty\_children}_i + \beta_2 \text{ fin\_satisfaction}_i + \beta_3$**

**$\text{age}_i + \beta_4 \text{ sex}_i + \beta_5 \text{ marital\_status}_i + \beta_6 \text{ urban\_rural}_i$**

**$+ \beta_7 \text{ GII}_i + (1 \mid \text{country}_i)$**

## Random Intercepts + Slopes for GII

|                    | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------------|----------|-----------|----------|----------|------|----------|----------|
| sd(Intercept)      | 2.08     | 0.58      | 1.29     | 3.58     | 1.00 | 351      | 374      |
| sd(GII)            | 4.71     | 1.13      | 2.95     | 7.52     | 1.00 | 404      | 385      |
| cor(Intercept,GII) | -0.68    | 0.25      | -0.98    | -0.06    | 1.01 | 262      | 317      |

|                  | Estimate     | Est.Error    | Q2.5         | Q97.5        |
|------------------|--------------|--------------|--------------|--------------|
| Intercept[1]     | -1.363750394 | 0.9238358629 | -3.063004902 | 0.474703317  |
| Intercept[2]     | -1.004541053 | 0.9239578916 | -2.703794523 | 0.837462871  |
| Intercept[3]     | -0.682135658 | 0.9241461059 | -2.379417292 | 1.159393484  |
| Intercept[4]     | -0.430508544 | 0.9242687268 | -2.125653560 | 1.415793169  |
| Intercept[5]     | 0.264751002  | 0.9242322169 | -1.440502097 | 2.101841459  |
| Intercept[6]     | 0.634043118  | 0.9241689674 | -1.070062326 | 2.469018169  |
| Intercept[7]     | 1.002021835  | 0.9244083626 | -0.693786843 | 2.836438241  |
| Intercept[8]     | 1.514835805  | 0.9243508177 | -0.189331934 | 3.338808323  |
| Intercept[9]     | 1.887445260  | 0.9242548114 | 0.180872270  | 3.719540173  |
| duty_children    | 0.312331581  | 0.0056589297 | 0.300984841  | 0.322471838  |
| fin_satisfaction | -0.008590692 | 0.0026645434 | -0.013648220 | -0.003653572 |
| age              | -0.006021103 | 0.0004228047 | -0.006866244 | -0.005232822 |
| sex              | -0.046987084 | 0.0132304702 | -0.072691923 | -0.020791146 |
| marital_status   | 0.028026089  | 0.0031065705 | 0.022132055  | 0.033892847  |
| urban_rural      | -0.179975232 | 0.0142390623 | -0.207543913 | -0.151353432 |
| GII              | -2.938204928 | 1.8038707090 | -6.367092059 | 0.503111668  |

$\text{abortion\_justif}_i \sim \text{OrderedLogistic}(\phi_i, \alpha)$

$\phi_i = \beta_1 \text{ duty\_children}_i + \beta_2 \text{ fin\_satisfaction}_i + \beta_3 \text{ age}_i + \beta_4 \text{ sex}_i$

$+ \beta_5 \text{ marital\_status}_i + \beta_6 \text{ urban\_rural}_i + \beta_7 \text{ GII}_i + (1 + \text{GII}_i \mid \text{country}_i)$

$\beta_j \sim \text{Normal}(0, 2)$

$\sigma_{\text{country}} \sim \text{Exponential}(1)$

$\alpha_k \sim \text{Normal}(0, 5)$

$\text{cor} \sim \text{LKJ}(2)$



## Random Intercepts + Slopes for GII

|                    | Estimate | Est.Error | l-95% CI | u-95% CI | Rhat | Bulk_ESS | Tail_ESS |
|--------------------|----------|-----------|----------|----------|------|----------|----------|
| sd(Intercept)      | 2.08     | 0.58      | 1.29     | 3.58     | 1.00 | 351      | 374      |
| sd(GII)            | 4.71     | 1.13      | 2.95     | 7.52     | 1.00 | 404      | 385      |
| cor(Intercept,GII) | -0.68    | 0.25      | -0.98    | -0.06    | 1.01 | 262      | 317      |

|                  | Estimate     | Est.Error    | Q2.5         | Q97.5        |
|------------------|--------------|--------------|--------------|--------------|
| Intercept[1]     | -1.363750394 | 0.9238358629 | -3.063004902 | 0.474703317  |
| Intercept[2]     | -1.004541053 | 0.9239578916 | -2.703794523 | 0.837462871  |
| Intercept[3]     | -0.682135658 | 0.9241461059 | -2.379417292 | 1.159393484  |
| Intercept[4]     | -0.430508544 | 0.9242687268 | -2.125653560 | 1.415793169  |
| Intercept[5]     | 0.264751002  | 0.9242322169 | -1.440502097 | 2.101841459  |
| Intercept[6]     | 0.634043118  | 0.9241689674 | -1.070062326 | 2.469018169  |
| Intercept[7]     | 1.002021835  | 0.9244083626 | -0.693786843 | 2.836438241  |
| Intercept[8]     | 1.514835805  | 0.9243508177 | -0.189331934 | 3.338808323  |
| Intercept[9]     | 1.887445260  | 0.9242548114 | 0.180872270  | 3.719540173  |
| duty_children    | 0.312331581  | 0.0056589297 | 0.300984841  | 0.322471838  |
| fin_satisfaction | -0.008590692 | 0.0026645434 | -0.013648220 | -0.003653572 |
| age              | -0.006021103 | 0.0004228047 | -0.006866244 | -0.005232822 |
| sex              | -0.046987084 | 0.0132304702 | -0.072691923 | -0.020791146 |
| marital_status   | 0.028026089  | 0.0031065705 | 0.022132055  | 0.033892847  |
| urban_rural      | -0.179975232 | 0.0142390623 | -0.207543913 | -0.151353432 |
| GII              | -2.938204928 | 1.8038707090 | -6.367092059 | 0.503111668  |

$\text{abortion\_justif}_i \sim \text{OrderedLogistic}(\phi_i, \alpha)$

$\phi_i = \beta_1 \text{ duty\_children}_i + \beta_2 \text{ fin\_satisfaction}_i + \beta_3 \text{ age}_i + \beta_4 \text{ sex}_i$

$+ \beta_5 \text{ marital\_status}_i + \beta_6 \text{ urban\_rural}_i + \beta_7 \text{ GII}_i + (1 + \text{GII}_i \mid \text{country}_i)$

$\beta_j \sim \text{Normal}(0, 2)$

$\sigma_{\text{country}} \sim \text{Exponential}(1)$

$\alpha_k \sim \text{Normal}(0, 5)$

$\text{cor} \sim \text{LKJ}(2)$

## CONCLUSION



- ❖ GII is consistently negatively associated with abortion justifiability in all models
- ❖ Fixed Effects Model: Strong, stable negative effect of GII.
- ❖ Random Slopes Model: GII varies significantly across countries
- ❖ In countries with higher baseline support, GII matters less.
- ❖ The association between GII and abortion attitudes is strong, yet context-dependent.